Mark Kahley

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EDUCATION

Texas A&M University College Station, Texas
Bachelor of Science in Aerospace Engineering; Minor in Mathematics (Expected May 2028)

- **3.86/4.0 GPA**
- Member of **Engineering Honors Program** (top 10%)
- **Technical Skills:** SolidWorks (Parametric CAD), MATLAB, Python, GitHub, OpenRocket, LaTeX, Excel

EXPERIENCE

Texas A&M Rocket Engine Design (RED) College Station, Texas
Forebody Team Member — Icarus Liquid Launch Vehicle (Sep 2026 – Present)

- Selected via technical interview for the Forebody team on Icarus, a student-built supersonic pressure-fed ethanol/N_2O liquid bipropellant rocket competing in the TCPF FAR challenge with an 85,000 ft target apogee
- Developing a from-scratch 6-DOF trajectory simulator in Python with quaternion rigid-body dynamics, RK4 integration, RASAero II aerodynamic tables, and `.eng` thrust curves; verified mass properties against OpenRocket
- Led an Excel airframe diameter trade study on apogee and stability, driving the adoption of a 7.5″ airframe

Students for the Exploration and Development of Space (SEDS) College Station, Texas
Officer — Student Astronaut Corps Coordinator (Apr 2026 – Present)

- Lead 20+ underclassmen in a semester-long, hands-on rocketry program, covering design, fabrication, and launch
- Coordinate the Student Astronaut Corps through technical workshops, guest speaker events, and tours

PROJECTS

3-DOF Rocket Flight Simulator & Monte Carlo Dispersion Analysis (Aug 2026)

- Developed a 3-DOF point-mass flight simulator in Python, driven by real imported motor thrust curves and a first-principles 3-layer ISA model
- Validated ascent performance to within 1% of OpenRocket and built ballistic, single, and dual deployment recovery modes via per-rocket YAML configuration files, decoupling physical parameters from simulation code
- Implemented a 1,000-run Monte Carlo dispersion analysis randomizing wind, launch angle, and drag coefficient to characterize landing footprint and apogee variance, applying range-safety-style statistical methodology

E-Class Model Rocket Design & Fabrication (May 2026)

- Engineered and simulated a custom E-class impulse rocket using SolidWorks and OpenRocket, analyzing flight stability to predict a 1,227 ft apogee
- Modeled and 3D-printed custom structural components, optimized for aerodynamic efficiency
- Executed the full design-build-test cycle, achieving a successful launch to 1,064 ft (15% error vs. simulation) and executing a nominal parachute deployment and recovery

Turbojet Engine Thermodynamic Analysis (Dec 2025)

- Modeled an ideal air-standard Brayton cycle for a Pratt & Whitney F-100 turbojet engine, executing thermodynamic state calculations across all core engine components and stages
- Calculated a net thrust at specified conditions within 10% of published reference values
- Generated temperature-entropy (T-s) and pressure-volume (P-v) diagrams in Excel, and documented analysis and results in a professionally formatted technical report

Mark's Real Estate Runners (May 2023 – Jan 2026)

- Founded and operated a courier/signage logistics and property service, serving 10+ luxury listings weekly

EXTRACURRICULAR ACTIVITIES

Alpha Sigma Phi Fraternity – Philanthropy Board Member (Mar 2025 – Present)
AIAA – Student Member (Sep 2025 – Present)